

International Cybersecurity and Digital Forensics Academy (ICDFA)

SBT-DF203 - Basic Networking Skills for Digital Forensics

LAB 4: SMTP Email Traffic Forensics

Forensic Investigation Report

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Evidence: smtp.pcap (supplied Wireshark sample capture)

1. Executive Summary

This report documents a controlled forensic examination of the supplied smtp.pcap training capture. " The analysis reconstructed an SMTP session between client 10.10.1.4:1470 and server 74.53.140.153:25. " The capture contains a server 220 service-ready banner identifying Exim 4.69, an EHLO greeting, AUTH LOGIN authentication, " MAIL FROM and RCPT TO envelope commands, and subsequent message data. The server advertised AUTH PLAIN/LOGIN and STARTTLS, " but the supplied evidence shows no client STARTTLS command; therefore the authentication exchange and SMTP application content " were visible in the captured plaintext stream.

The recovered message headers identify the sender as Gurpartap Singh <gurpartap@patriots.in>, the recipient as " raj_deol2002in@yahoo.co.in, the subject as SMTP, and Microsoft Office Outlook 12.0 as the mail client. " The message is identified as multipart/mixed with multipart/alternative and text/plain/text/html content types. " Sensitive authentication values were decoded offline for the lab, but only masked forms are reproduced in this public-facing report.

2. Authority, Scope and Safety Controls

The examination follows the SBT-DF203 Lab 4 procedure and uses only the supplied training capture. " No third-party system was targeted during analysis. The original capture was preserved, a working copy was created, " and SHA-256 values were recorded. Recovered credentials are treated as confidential assessment evidence and are masked in this report.

3. Evidence Description and Integrity

Evidence Item	Finding
Original capture	evidence/smtp.pcap
Working copy	working/smtp_working.pcap
Original SHA-256	17ad230db1b6fd5dd18eb311092dfc1f6eb162054bdb47697b89bef5a86a47ab
Working-copy SHA-256	17ad230db1b6fd5dd18eb311092dfc1f6eb162054bdb47697b89bef5a86a47ab
Packet count	60
Capture duration	9.198384 seconds
Earliest packet	2009-10-05 02:06:07.492060 -0400
Latest packet	2009-10-05 02:06:16.690444 -0400
Encapsulation	Ethernet / pcap

4. SMTP Timeline

Time	Direction	Command/Code	Meaning
02:06:08.219663	Server → Client	220	Service ready; Exim 4.69 banner.
02:06:08.224809	Client → Server	EHLO GP	Client greeting.
02:06:08.566183	Server → Client	250	Capabilities include AUTH PLAIN LOGIN and STARTTLS.

02:06:08.568729	Client → Server	AUTH LOGIN	Authentication method selected.
02:06:08.911081	Server → Client	334	Username prompt encoded in Base64.
02:06:08.911655	Client → Server	Base64 username	Authentication username transmitted.
02:06:09.253544	Server → Client	334	Password prompt encoded in Base64.
02:06:09.254118	Client → Server	Base64 password	Authentication password transmitted.
02:06:09.613798	Server → Client	235	Authentication succeeded.
02:06:09.614414	Client → Server	MAIL FROM	Envelope sender accepted for processing.
02:06:09.957250	Client → Server	RCPT TO	Envelope recipient accepted.
02:06:10.320203	Client → Server	DATA	Message content phase begins.

5. Command and Response Analysis

The SMTP exchange follows the expected application sequence: server greeting (220), client EHLO, capability response (250), " AUTH LOGIN with two 334 challenges, successful authentication (235), envelope sender and recipient commands, and DATA. " The capture therefore provides sufficient application-layer evidence to reconstruct the SMTP session and relevant message headers.

6. Authentication Evidence and Redaction

The AUTH LOGIN exchange contains Base64-encoded username and password fields. Base64 is an encoding rather than encryption, " so the values were decoded offline using Python's base64 module as prescribed by the lab. The decoded values are not reproduced in full " because the lab requires sensitive credentials to be masked in the report.

Field	Masked finding
Username	g*****@patriots.in
Password	px*****23
Authentication result	235 Authentication succeeded

7. Email Reconstruction

Header / attribute	Recovered value
From	Gurpartap Singh <gurpartap@patriots.in>
To	<raj_deol2002in@yahoo.co.in>
Subject	SMTP
Date	Mon, 5 Oct 2009 11:36:07 +0530
Message-ID	<000301ca4581\$ef9e57f0\$cedb07d0\$Min> (as captured/OCR-visible)
MIME-Version	1.0

Content-Type	multipart/mixed;
Nested content types	multipart/alternative; text/plain; text/html; text/plain
X-Mailer	Microsoft Office Outlook 12.0

The message is multipart/mixed and contains alternative/plain-text and HTML representations. " The supplied screenshot evidence does not provide a reliable attachment filename or a complete attachment-object extraction, " so no specific attachment is asserted as recovered. The body content should therefore be treated as only partially documented from the supplied screenshots.

8. Client, Server and Network Metadata

Metadata	Finding
Client IP / port	10.10.1.4 / TCP 1470
Server IP / port	74.53.140.153 / TCP 25
Client MAC	00:01:33:d9:81:60
Server MAC	00:e0:1c:3c:17:c2
SMTP server	xc90.websiteswelcome.com
SMTP software	Exim 4.69
Client software	Microsoft Office Outlook 12.0
EHLO-associated address reported by server	122.162.143.157

9. Encryption / STARTTLS Assessment

The server capability response advertises STARTTLS, but the supplied capture shows no client STARTTLS request. " The AUTH LOGIN credentials and message headers are therefore visible in the captured SMTP stream. This finding is based on the " observed application-layer commands; the presence of port 25 alone is not used to infer encryption.

Because the capture does not show a TLS transition, there is no basis in this evidence set to claim that the SMTP application " payload after authentication was protected by TLS. The visible plaintext exchange is the principal evidential limitation and security finding.

10. Evidential Limitations

- The analysis is based on the supplied historical training PCAP and screenshot evidence.
- The screenshots do not establish a complete attachment filename/object extraction; multipart/mixed is observed.
- Sensitive credentials were visible in the source evidence but are intentionally masked in this report.
- The displayed Message-ID contains OCR-visible ambiguity; the value should be verified directly from the original PCAP before formal external use.
- MAC addresses and endpoint metadata are reported from the supplied tshark output and should be interpreted within the capture's historical context.

11. Evidence Index and Screenshot Log

Part A - Evidence Preparation

```
kali@kali: ~/SBT-DF203-Lab4
Session Actions Edit View Help
[sudo] password for kali: 
(kali@kali)-[~]
$ mkdir -p ~/SBT-DF203-Lab4/{evidence,working,exported,reports,screenshots,scripts}

(kali@kali)-[~]
$ cd ~/SBT-DF203-Lab4

(kali@kali)-[~/SBT-DF203-Lab4]
$ sudo apt update && sudo apt install -y wireshark tshark python3
Hit:1 http://http.kali.org/kali kali-rolling InRelease
Hit:2 https://repo.protonvpn.com/debian stable InRelease
1822 packages can be upgraded. Run 'apt list --upgradable' to see them.
wireshark is already the newest version (4.6.6-1).
tshark is already the newest version (4.6.6-1).
python3 is already the newest version (3.14.6-1).
python3 set to manually installed.
The following packages were automatically installed and are no longer required:
 libimath-3-1-29t64 libpython3.13-dev python3.13 rpcsvc-proto
 libmbdcrypto16 libraw23t64 python3.13-dev
 libpostal-data libvidstab1.1 python3.13-minimal
 libpostal1 libzxing3 python3.13-venv
Use 'sudo apt autoremove' to remove them.

Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 1822
```

Figure 1: Lab workspace and required forensic tools verified.

```
(kali@kali)-[~/SBT-DF203-Lab4]
$ wget -O evidence/smtp.pcap 'https://wiki.wireshark.org/uploads/__moin_import__/_attachments/SampleCaptures/smtp.pcap'
--2026-09-13 20:18:25-- https://wiki.wireshark.org/uploads/__moin_import__/_attachments/SampleCaptures/smtp.pcap
Resolving wiki.wireshark.org (wiki.wireshark.org) ... 172.67.75.39, 104.26.11.240, 104.26.10.240, ...
Connecting to wiki.wireshark.org (wiki.wireshark.org)|172.67.75.39|:443 ... connected.
HTTP request sent, awaiting response ... 302 Found
Location: https://gitlab.com/wireshark/wireshark/-/wikis/uploads/__moin_import__/_attachments/SampleCaptures/smtp.pcap [following]
--2026-09-13 20:18:30-- https://gitlab.com/wireshark/wireshark/-/wikis/uploads/__moin_import__/_attachments/SampleCaptures/smtp.pcap
Resolving gitlab.com (gitlab.com) ... 172.65.251.78, 2606:4700:90:0:f22e:fbec:5be:a9b9
Connecting to gitlab.com (gitlab.com)|172.65.251.78|:443 ... connected.
HTTP request sent, awaiting response ... 200 OK
Length: 27850 (27K) [application/octet-stream]
Saving to: 'evidence/smtp.pcap'

evidence/smtp.pcap  100%[=====>] 27.20K  --.-KB/s   in 0.02s

2026-09-13 20:18:31 (1.33 MB/s) - 'evidence/smtp.pcap' saved [27850/27850]
```

Figure 2: SMTP sample capture downloaded into the evidence directory.

```
(kali㉿kali)-[~/SBT-DF203-Lab4]
$ cp --preserve=timestamps evidence/smtp.pcap working/smtp_working.pcap

(kali㉿kali)-[~/SBT-DF203-Lab4]
$ sha256sum evidence/smtp.pcap working/smtp_working.pcap | tee reports/smtp_capture_hashes.txt
17ad230db1b6fd5dd18eb311092df1cf6eb162054bdb47697b89bef5a86a47ab  evidence/smtp.pcap
17ad230db1b6fd5dd18eb311092df1cf6eb162054bdb47697b89bef5a86a47ab  working/smtp_working.pcap

(kali㉿kali)-[~/SBT-DF203-Lab4]
$ ss
```

Figure 3: Original and working-copy SHA-256 hashes match.

```
Session  Actions  Edit  View  Help
(kali㉿kali)-[~/SBT-DF203-Lab4]
$ capinfos evidence/smtp.pcap | tee reports/smtp_capinfos.txt
File name:          evidence/smtp.pcap
File type:          Wireshark/tcpdump/... - pcap
File encapsulation: Ethernet
File timestamp precision: microseconds (6)
Packet size limit:  file hdr: 65535 bytes
Number of packets:  60
File size:          27 kB
Data size:          26 kB
Capture duration:   9.198384 seconds
Earliest packet time: 2009-10-05 02:06:07.492060
Latest packet time:  2009-10-05 02:06:16.690444
Data byte rate:     2,920 bytes/s
Data bit rate:      23 kbps
Average packet size: 447.77 bytes
Average packet rate: 6 packets/s
SHA256:             17ad230db1b6fd5dd18eb311092df1cf6eb162054bdb47697b89bef5a86a47ab
SHA1:               3def1a1fed849e7f23b66e6925ddff05845a400b
Strict time order:  True
Number of interfaces in file: 1
Interface #0 info:
    Encapsulation = Ethernet (1 - ether)
    Capture length = 65535
    Time precision = microseconds (6)
    Time ticks per second = 1000000
    Number of stat entries = 0
    Number of packets = 60

(kali㉿kali)-[~/SBT-DF203-Lab4]
```

Figure 4: capinfos summary for smtp.pcap.

Part B - SMTP Stream and Command Analysis

```

kali@kali: ~/SBT-DF203-Lab4
Session Actions Edit View Help
(kali@kali)-[~/SBT-DF203-Lab4]
$ tshark -r working/smtp_working.pcap -q -z conv,tcp | tee reports/tcp_conversations.txt

TCP Conversations
Filter:<No Filter>

→ | | Total | Relative | Duration | ← | |
  | |      | |      |          |   | |
Frames Bytes | | Frames Bytes | Start | Frames Bytes | | F
10.10.1.4:1470 ↔ 74.53.140.153:25 0.036986000 7.5777
28 22 kB 53 24 kB

(kali@kali)-[~/SBT-DF203-Lab4]
$

```

Figure 5: TCP conversation inventory identifying the SMTP client/server pair.

```

kali@kali: ~/SBT-DF203-Lab4
Session Actions Edit View Help
(kali@kali)-[~/SBT-DF203-Lab4]
$ tshark -r working/smtp_working.pcap -Y 'smtp' -T fields \
> -e frame.number -e frame.time -e ip.src -e tcp.srcport -e ip.dst -e tcp.dstport -e _ws.col.Info \
> | tee reports/smtp_packet_inventory.tsv
6      2009-10-05T02:06:08.219663000-0400      74.53.140.153      25      10.10.1.
4      1470      S: 220-xc90.websitewelcome.com ESMTP Exim 4.69 #1 Mon, 05 Oct 20
09 01:05:54 -0500 | We do not authorize the use of this system to transport uns
olicited, | and/or bulk e-mail.
7      2009-10-05T02:06:08.224809000-0400      10.10.1.4      1470      74.53.14
0.153      25      C: EHLO GP
9      2009-10-05T02:06:08.566183000-0400      74.53.140.153      25      10.10.1.
4      1470      S: 250-xc90.websitewelcome.com Hello GP [122.162.143.157] | SIZE
52428800 | PIPELINING | AUTH PLAIN LOGIN | STARTTLS | HELP
10     2009-10-05T02:06:08.568729000-0400      10.10.1.4      1470      74.53.14
0.153      25      C: AUTH LOGIN
11     2009-10-05T02:06:08.911081000-0400      74.53.140.153      25      10.10.1.
4      1470      S: 334 VXNlcm5hbWU6
12     2009-10-05T02:06:08.911655000-0400      10.10.1.4      1470      74.53.14
0.153      25      C: User: Z3VycGFydGFwQHhhdHJpb3RzLmlu
13     2009-10-05T02:06:09.253544000-0400      74.53.140.153      25      10.10.1.
4      1470      S: 334 UGFzc3dvcmQ6
14     2009-10-05T02:06:09.254118000-0400      10.10.1.4      1470      74.53.14
0.153      25      C: Pass: cHVuamFiQDEyMw==
15     2009-10-05T02:06:09.613798000-0400      74.53.140.153      25      10.10.1.
4      1470      S: 235 Authentication succeeded
16     2009-10-05T02:06:09.614414000-0400      10.10.1.4      1470      74.53.14
0.153      25      C: MAIL FROM: <gurpartap@patriots.in>
17     2009-10-05T02:06:09.956765000-0400      74.53.140.153      25      10.10.1.
4      1470      S: 250 OK
18     2009-10-05T02:06:09.957250000-0400      10.10.1.4      1470      74.53.14
0.153      25      C: RCPT TO: <raj_deol2002in@yahoo.co.in>

```

Figure 6: SMTP packet inventory showing server banner, EHLO, AUTH LOGIN and envelope activity.

```

Session Actions Edit View Help
└─$ tshark -r working/smtp_working.pcap -Y 'smtp.req || smtp.rsp' -T fields -e f
rame.number -e frame.time -e ip.src -e ip.dst -e smtp.req.command -e smtp.req.pa
rameter -e smtp.response.code -e smtp.rsp.parameter | tee reports/smtp_commands_
responses.tsv
6      2009-10-05T02:06:08.219663000-0400      74.53.140.153      10.10.1.4      2
20      xc90.websitewelcome.com ESMTP Exim 4.69 #1 Mon, 05 Oct 2009 01:05:54 -05
00 ,We do not authorize the use of this system to transport unsolicited, ,and/or
bulk e-mail.
7      2009-10-05T02:06:08.224809000-0400      10.10.1.4      74.53.140.153      E
HLO      GP
9      2009-10-05T02:06:08.566183000-0400      74.53.140.153      10.10.1.4      2
50      xc90.websitewelcome.com Hello GP [122.162.143.157],SIZE 52428800,PIPELIN
ING,AUTH PLAIN LOGIN,STARTTLS,HELP
10     2009-10-05T02:06:08.568729000-0400      10.10.1.4      74.53.140.153      A
UTH      LOGIN
11     2009-10-05T02:06:08.911081000-0400      74.53.140.153      10.10.1.4      3
34     VXNlcm5hbWU6
12     2009-10-05T02:06:08.911655000-0400      10.10.1.4      74.53.140.153
13     2009-10-05T02:06:09.253544000-0400      74.53.140.153      10.10.1.4      3
34     UGFzc3dvcmQ6
14     2009-10-05T02:06:09.254118000-0400      10.10.1.4      74.53.140.153
15     2009-10-05T02:06:09.613798000-0400      74.53.140.153      10.10.1.4      2
35     Authentication succeeded
16     2009-10-05T02:06:09.614414000-0400      10.10.1.4      74.53.140.153      M
AIL      FROM: <gurpartap@patriots.in>
17     2009-10-05T02:06:09.956765000-0400      74.53.140.153      10.10.1.4      2
50     OK
18     2009-10-05T02:06:09.957250000-0400      10.10.1.4      74.53.140.153      R
CPT      TO: <raj_deol2002in@yahoo.co.in>
19     2009-10-05T02:06:10.319708000-0400      74.53.140.153      10.10.1.4      2
50     Accepted
20     2009-10-05T02:06:10.320203000-0400      10.10.1.4      74.53.140.153      D

```

Figure 7: SMTP command/response extraction, including 220, 250, 334 and 235 responses.

Part C - Authentication Evidence and Redaction

```

(kali@kali)-[~/SBT-DF203-Lab4]
└─$ nano scripts/decode_auth.py

(kali@kali)-[~/SBT-DF203-Lab4]
└─$ python3 scripts/decode_auth.py | tee reports/decoded_auth_full.txt
username: gurpartap@patriots.in
password: punjab@123

(kali@kali)-[~/SBT-DF203-Lab4]
└─$

```

```

kali@kali: ~/SBT-DF203-Lab4
Session Actions Edit View Help
GNU nano 8.7.1 scripts/decode_auth.py *
import base64

samples = {
    'username': 'Z3VycGFydGFwQHBhdHJpb3RzLm1u',
    'password': 'cHVuamFiQDEyMw=='
}

for label, value in samples.items():
    try:
        decoded = base64.b64decode(value).decode('utf-8', errors='replace')
        print(f'{label}: {decoded}')
    except Exception as exc:
        print(f'{label}: decode failed: {exc}')

```

Figure 9: Offline Base64 decoding output; sensitive values are masked in this report.

```

Session Actions Edit View Help
(kali@kali)-[~/SBT-DF203-Lab4]
$ echo "username: g*****@patriots.in" > reports/decoded_auth_masked.txt

(kali@kali)-[~/SBT-DF203-Lab4]
$ echo "password: p*****23" >> reports/decoded_auth_masked.txt

(kali@kali)-[~/SBT-DF203-Lab4]
$ cat reports/decoded_auth_masked.txt
username: g*****@patriots.in
password: p*****23

(kali@kali)-[~/SBT-DF203-Lab4]
$

```

Figure 10: Masked authentication evidence prepared for reporting.

Part D - Email Reconstruction

```
(kali@kali)-[~/SBT-DF203-Lab4]
$ grep -Ei '^([Date|From|To|Subject|Message-ID|MIME-Version|Content-Type|User-Agent|X-Mailer]):' \
reports/smtp_stream_0.txt | tee reports/message_headers.txt
From: "Gurpartap Singh" <gurpartap@patriots.in>
To: <raj_deol2002in@yahoo.co.in>
Subject: SMTP
Date: Mon, 5 Oct 2009 11:36:07 +0530
Message-ID: <000301ca4581$ef9e57f0$cedb07d0$@in>
MIME-Version: 1.0
Content-Type: multipart/mixed;
X-Mailer: Microsoft Office Outlook 12.0
Content-Type: multipart/alternative;
Content-Type: text/plain;
Content-Type: text/html;
Content-Type: text/plain;

(kali@kali)-[~/SBT-DF203-Lab4]
$
```

Figure 12: Recovered message headers and Microsoft Outlook 12.0 client indicator.

```
(kali@kali)-[~/SBT-DF203-Lab4]
$ sed -e 's/Z3VycGFydGFwQHBhdHJpb3RzLmlu/[REDACTED-BASE64-USERNAME]/' \
> -e 's/cHVuamFiQDEyMw==/[REDACTED-BASE64-PASSWORD]/' \
> reports/smtp_stream_0.txt > reports/reconstructed_email_redacted.txt
```

```
Session Actions Edit View Help
(kali@kali)-[~/SBT-DF203-Lab4]
$ cat reports/reconstructed_email_redacted.txt | head -40

Follow: tcp,ascii
Filter: tcp.stream eq 0
Node 0: 10.10.1.4:1470
Node 1: 74.53.140.153:25
181
220-xc90.websitewelcome.com ESMTP Exim 4.69 #1 Mon, 05 Oct 2009 01:05:54 -0500
220-We do not authorize the use of this system to transport unsolicited,
220 and/or bulk e-mail.

9
EHLO GP

137
250-xc90.websitewelcome.com Hello GP [122.162.143.157]
250-SIZE 52428800
250-PIPELINING
250-AUTH PLAIN LOGIN
250-STARTTLS
250 HELP

12
AUTH LOGIN

18
334 VXNlcm5hbWU6

30
[REDACTED-BASE64-USERNAME]
```

```
137
250-xc90.websitewelcome.com Hello GP [122.162.143.157]
250-SIZE 52428800
250-PIPELINING
250-AUTH PLAIN LOGIN
250-STARTTLS
250 HELP

12
AUTH LOGIN

18
334 VXNlcm5hbWU6

30
[REDACTED-BASE64-USERNAME]

18
334 UGFzc3dvcmQ6

18
[REDACTED-BASE64-PASSWORD]

30
235 Authentication succeeded

(kali@kali)-[~/SBT-DF203-Lab4]
$
```

Figure 14: Redacted Follow TCP Stream reconstruction.

Part E - Network Metadata

```

Session Actions Edit View Help
(kali@kali)-[~/SBT-DF203-Lab4]
$ tshark -r working/smtp_working.pcap -Y 'smtp' -T fields \
> -e frame.number -e frame.time_epoch -e eth.src -e eth.dst -e ip.src -e tcp.src
port \
> -e ip.dst -e tcp.dstport -e tcp.stream \
> | tee reports/smtp_network_metadata.tsv
6      1254722768.219663000      00:1f:33:d9:81:60      00:e0:1c:3c:17:c2      7
4.53.140.153      25      10.10.1.4      1470      0
7      1254722768.224809000      00:e0:1c:3c:17:c2      00:1f:33:d9:81:60      1
0.10.1.4      1470      74.53.140.153      25      0
9      1254722768.566183000      00:1f:33:d9:81:60      00:e0:1c:3c:17:c2      7
4.53.140.153      25      10.10.1.4      1470      0
10     1254722768.568729000      00:e0:1c:3c:17:c2      00:1f:33:d9:81:60      1
0.10.1.4      1470      74.53.140.153      25      0
11     1254722768.911081000      00:1f:33:d9:81:60      00:e0:1c:3c:17:c2      7
4.53.140.153      25      10.10.1.4      1470      0
12     1254722768.911655000      00:e0:1c:3c:17:c2      00:1f:33:d9:81:60      1
0.10.1.4      1470      74.53.140.153      25      0
13     1254722769.253544000      00:1f:33:d9:81:60      00:e0:1c:3c:17:c2      7
4.53.140.153      25      10.10.1.4      1470      0
14     1254722769.254118000      00:e0:1c:3c:17:c2      00:1f:33:d9:81:60      1
0.10.1.4      1470      74.53.140.153      25      0
15     1254722769.613798000      00:1f:33:d9:81:60      00:e0:1c:3c:17:c2      7
4.53.140.153      25      10.10.1.4      1470      0
16     1254722769.614414000      00:e0:1c:3c:17:c2      00:1f:33:d9:81:60      1
0.10.1.4      1470      74.53.140.153      25      0
17     1254722769.956765000      00:1f:33:d9:81:60      00:e0:1c:3c:17:c2      7
4.53.140.153      25      10.10.1.4      1470      0
18     1254722769.957250000      00:e0:1c:3c:17:c2      00:1f:33:d9:81:60      1
0.10.1.4      1470      74.53.140.153      25      0
19     1254722770.319708000      00:1f:33:d9:81:60      00:e0:1c:3c:17:c2      7
4.53.140.153      25      10.10.1.4      1470      0

(kali@kali)-[~/SBT-DF203-Lab4]
$ tshark -r working/smtp_working.pcap -Y 'smtp contains "User-Agent" || smtp c
ontains "X-Mailer"' \
> -T fields -e frame.number -e tcp.stream -e smtp.req.parameter -e data-text-lin
es \
> | tee reports/client_indicators.tsv
22      0      Line-based text data (47 lines)
26      0      Line-based text data (12 lines)

(kali@kali)-[~/SBT-DF203-Lab4]
$

```

Figure 16: Network metadata showing IP addresses, ports and MAC addresses.

Part F - Encryption and STARTTLS Assessment

```
(kali@kali)-[~/SBT-DF203-Lab4]
$ tshark -r working/smtp_working.pcap -Y 'smtp.rsp.parameter contains "STARTTLS"' \
> -T fields -e frame.number -e frame.time -e smtp.rsp.parameter \
> | tee reports/starttls_check.txt
9      2009-10-05T02:06:08.566183000-0400      xc90.websitewelcome.com Hello GP
[122.162.143.157],SIZE 52428800,PIPELINING,AUTH PLAIN LOGIN,STARTTLS,HELP

(kali@kali)-[~/SBT-DF203-Lab4]
$ cat reports/starttls_check.txt
9      2009-10-05T02:06:08.566183000-0400      xc90.websitewelcome.com Hello GP
[122.162.143.157],SIZE 52428800,PIPELINING,AUTH PLAIN LOGIN,STARTTLS,HELP

(kali@kali)-[~/SBT-DF203-Lab4]
$
```

Figure 18: STARTTLS capability advertised by the SMTP server.

```
(kali@kali)-[~/SBT-DF203-Lab4]
$ tshark -r working/smtp_working.pcap -Y 'smtp.req.command = "STARTTLS"' \
> | tee reports/starttls_invoked_check.txt

(kali@kali)-[~/SBT-DF203-Lab4]
$ cat reports/starttls_invoked_check.txt

(kali@kali)-[~/SBT-DF203-Lab4]
$
```

Figure 19: STARTTLS invocation check showing no STARTTLS request in the capture.

12. Conclusions

The supplied evidence supports reconstruction of a plaintext SMTP session from 10.10.1.4:1470 to 74.53.140.153:25. " The server identified itself as Exim 4.69 and offered AUTH PLAIN/LOGIN and STARTTLS. The client selected AUTH LOGIN, " completed authentication successfully, and submitted a message using MAIL FROM, RCPT TO and DATA. Message headers identify " Microsoft Office Outlook 12.0 and a multipart/mixed email with alternative plain-text/HTML content. No STARTTLS invocation is " shown in the supplied capture, so the authentication and application-layer SMTP evidence remained visible in the analyzed stream.

13. Student Declaration

I confirm that I completed this laboratory exercise in an authorized training environment, preserved the supplied evidence, used a working copy for analysis, protected sensitive authentication data through redaction, and documented findings from the available evidence.